Trigonometry and Modelling 1

Pure Mathematics Year 2

Chapter 7

Trigonometry and Modelling

Chapter Overview 2

This chapter develops addition and double-angle formulae, combines sine and cosine into one function, and uses these ideas to solve equations and model periodic motion.

1. Addition formulae

$\sin(a+b) = \sin a\cos b + \cos a\sin b$

2. Double-angle formulae

Solve $2\tan 2y \tan y = 3$, $0 \le y < 2\pi$.

3. Combining sine and cosine

Find the maximum of $2\sin x + \cos x$ and where it occurs.

4. Modelling

A tide can be modelled by $x = R\sin(2\pi t/5 + \alpha)$, where t is hours after midnight.

Addition Formulae 3

Addition formulae handle sums and differences of angles.

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Do I need to memorise these?

They are in the formula booklet, but aim to memorise them, especially sine and cosine.

How to remember

The first function on the right matches the left. Sine mixes sine and cosine; cosine keeps matching pairs. Sine keeps the sign, cosine changes it. Tangent keeps the sign on top and changes it underneath.

The identities apply wherever both sides are defined.

A Common Error 4

Why is $\sin(A+B)$ not generally $\sin A + \sin B$? Give a counterexample.

Sine is a function, so it need not distribute over addition; similarly $(a+b)^2$ is not generally $a^2 + b^2$.

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1
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Recall the Addition Formulae 5

Complete $\sin(A+B)$.

1

Complete $\sin(A-B)$.

2

Complete $\cos(A+B)$.

3

Complete $\cos(A-B)$.

4

Complete $\tan(A+B)$.

5

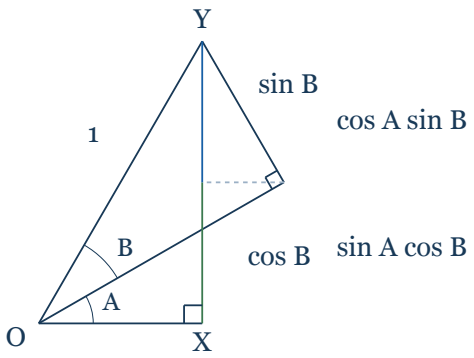
Complete $\tan(A-B)$.

6

How to remember: the first function on the right matches the left. Sine mixes the functions and keeps the sign; cosine pairs matching functions and changes the sign. Tangent keeps the sign above and changes it below.

Geometric Proof of the Sine Addition Formula 6

Geometric proof (not required for the exam). Let $OY = 1$ make angle $A+B$ with the horizontal, so $XY = \sin(A+B)$.



1. Resolve the unit line along directions making angles A and $A+\pi/2$ with the horizontal. The perpendicular components have lengths $\cos B$ and $\sin B$.
2. Their vertical components are $\sin A \cos B$ and $\cos A \sin B$.
3. Add the vertical lengths:
   $\sin(A+B) = \sin A \cos B + \cos A \sin B$.

Proof of other identities 7

Use the sine addition formula to prove the formula for $\sin(A - B)$.

1

Prove the formula for $\cos(A + B)$. Hint: use the complementary-angle relation.

2

Proof of other identities 8

Edexcel C3 January 2012, Q8(a).

Use the sine and cosine addition formulae to prove the tangent addition formula.

8(a)

Examples 9

Given $2\sin(x+y) = 3\cos(x-y)$, express $\tan x$ in terms of $\tan y$.

1

Exercises 7A – Pure Y2 P169 10

2. Use the sine and cosine subtraction formulae to prove the tangent subtraction formula.

2

3. Substitute $A = P, B = -Q$ into the sine addition formula to obtain $\sin(P-Q)$.

3

7. Write $\sin(x+\pi/6) = p\sin x + q\cos x$. Find p, q.

7

11. Given $\cos y = \sin(x + y)$, show that $\tan y = \sec x - \tan x$.

11

15. Given $\tan(x + \pi/3) = 1/2$, show that $\tan x = 8 - 5\sqrt{3}$.

15

16. Prove $\cos\theta + \cos(\theta + 2\pi/3) + \cos(\theta + 4\pi/3) = 0$. Show every stage of working.

16

Uses of Addition Formulae 13

Using a suitable angle formula, show $\sin 15^\circ = (\sqrt{6} - \sqrt{2})/4$.

1

Given $\sin A = -3/5$, $180^\circ < A < 270^\circ$, and $\cos B = -12/13$ with B obtuse, find (a) $\cos(A - B)$, (b) $\tan(A + B)$.

Tip: $\cos A = \pm\sqrt{1 - \sin^2 A}$. Choose the sign from the quadrant.

a

Continued... 14

Given that $\sin A = -\frac{3}{5}$ and $180^\circ < A < 270^\circ$, and that $\cos B = -\frac{12}{13}$, and that B is obtuse, find the value of: (b) $\tan(A + B)$

b

Test Your Understanding 15

Without a calculator, find $\cos 75°$.

a

Without a calculator, find $\tan 75°$.

b

Challenging question 16

Without a calculator. Edexcel June 2013 Q3.

a. Given $2\cos(x + 50°) = \sin(x + 40°)$, prove $\tan x = \frac{1}{3}\tan 40°$.

a

b. Solve $2\cos(2\theta + 50°) = \sin(2\theta + 40°)$, $0 \leq \theta < 360°$.

b

3. a. Use the tangent addition formula for $\tan(45^\circ + 30^\circ)$.
   b. Hence show $\tan 75^\circ = 2 + \sqrt{3}$.

3a

3b

5. a. Show $\cos 105^\circ = (\sqrt{2} - \sqrt{6})/4$.
   b. Write $\sec 105^\circ = -\sqrt{a}(1 + \sqrt{b})$, finding integers a, b.

5a

5b

8. $\sin A = 8/17$, A acute; $\cos B = -4/5$, B obtuse. Find (a) $\sin(A - B)$, (b) $\cos(A - B)$, (c) $\cot(A - B)$.

General observations: use the quadrants to select the signs of the missing ratios.

a

b

c

Exercise 7B - Pure Y2 P172-173 19

Without a calculator. 10. $\tan A = 1/5$, $\tan B = 2/3$. Find $A + B$ when (a) both are acute, (b) A is reflex and B acute.

a

b

Extension: Difference Formulae 20

STEP I 2010 Q3.

a. Show

$$\sin(x+y) - \sin(x-y) = 2\cos x \sin y.$$

Hence show

$$\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}.$$

a

b. Show

$$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}.$$

b

c. Points P, Q, R, S have coordinates $(a\cos u, b\sin u)$ at $u = p, q, r, s$, with $a, b > 0$ and $0 \le p < q < r < s < 2\pi$. Assuming PQ, SR are not vertical, prove they are parallel exactly when $r + s - p - q = 2\pi$.

c

Double Angle Formulae 21

$$\sin 2A = 2\sin A\cos A$$
$$\cos 2A = \cos^2 A - \sin^2 A$$
$$= 2\cos^2 A - 1 = 1 - 2\sin^2 A$$
$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

Tip: in the cosine identities, the cosine term is positive and the sine term negative. The first cosine form is used less often.

Set $B = A$ in the addition formulae. For example:

$$\sin 2A = \sin(A + A)$$
$$= \sin A\cos A + \cos A\sin A = 2\sin A\cos A.$$

Double-angle formulae relate an angle to half that angle. Use them in either direction, where defined.

Examples 22

a. Write $\cos^2 50^\circ - \sin^2 50^\circ$ as one ratio.

a

b. Simplify $\dfrac{2\tan(\pi/6)}{1-\tan^2(\pi/6)}$.

b

c. Simplify $\dfrac{4\sin 70^\circ}{\sec 70^\circ}$.

c

Examples 23

Given $x = 3\sin\theta$ and $y = 3 - 4\cos 2\theta$, eliminate θ.

1

This changes parametric equations into a Cartesian equation, as in the next chapter.

Given $\cos x = 3/4$ and x acute, find (a) $\sin 2x$, (b) $\tan 2x$.

a

b

Exercise 7C – Pure Y2 p175-177 24

3. Use $\tan(A + A)$ to express $\tan 2A$ in terms of $\tan A$.

3

6. a. Show $(\sin A + \cos A)^2 = 1 + \sin 2A$.
   b. Hence find $(\sin(\pi/8) + \cos(\pi/8))^2$.

6a

6b

8. $p = 2\cos\theta, q = \cos 2\theta$. Express q in terms of p.

8

11. Find the possible values of $\sin\theta$ if $\cos 2\theta = 23/25$.

11

14. Given $\pi < \theta < 3\pi/2$ and $\tan\theta = 3/4$, find $\tan(\theta/2)$.

14

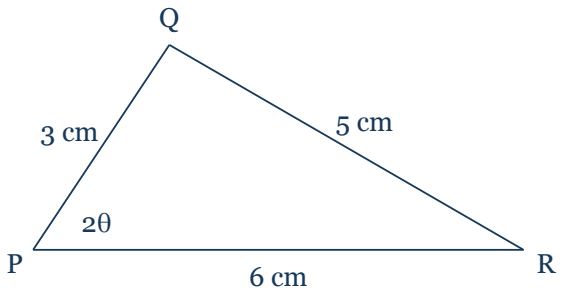
15. Given $\cos x + \sin x = m$, $\cos x - \sin x = n$, express $\cos 2x$ in terms of m, n.

15

16. In triangle PQR, $PQ = 3$, $PR = 6$, $QR = 5$ cm and $\angle QPR = 2\theta$.

a. Use the cosine rule to show $\cos 2\theta = 5/9$.

b. Find the exact value of $\sin\theta$.



16a

16b

Solving Trigonometric Equations 27

This is effectively the same type of question you encountered in Chapter 6 and in Year 1, except you may need to use either the addition formulae or double angle formulae.

[Textbook] Solve $3\cos 2x - \cos x + 2 = 0$ for $0 \leq x \leq 360°$.

1

Further Examples 28

a. Show $\sin 3A = 3\sin A - 4\sin^3 A$.

b. Hence solve $16\sin^3\theta - 12\sin\theta - 2\sqrt{3} = 0$, $0 < \theta < 2\pi$.

a

b

Solve $4\cos(\theta - 30°) = 8\sqrt{2}\sin\theta$, $0 \leq \theta < 360°$.

2

Test Your Understanding 29

Edexcel C3 January 2013 Q6.

i. Without a calculator, find exactly $(\sin 22.5^\circ + \cos 22.5^\circ)^2$, showing working.

ii. Show $\cos 2\theta + \sin\theta = 1$ becomes $k\sin^2\theta - \sin\theta = 0$, stating k. Hence solve for $0 \leq \theta < 360^\circ$.

i

ii

If you finish: solve $2\tan 2y \tan y = 3$, $0 \leq y < 2\pi$, to 2 d.p.

Tip: remember both signs when taking a square root.

Extra

Exercise 7D – Pure Y2 P180-181 30

2. a. Show $\sin(\theta + \pi/4) = (\sin\theta + \cos\theta)/\sqrt{2}$.

b. Hence solve $(\sin\theta + \cos\theta)/\sqrt{2} = 1/\sqrt{2}$, $0 \leq \theta \leq 2\pi$.

c. Deduce the solutions of $\sin\theta + \cos\theta = 1$ in this interval.

a

b

c

6. In triangle ABC, $AB = 4$ cm, $AC = 5$ cm, $\angle ABC = 2\theta$, $\angle ACB = \theta$. Find θ to 1 d.p.

6 diagram

6

7. a. Write $5\sin 2\theta + 4\sin\theta = 0$ as $a\sin\theta(b\cos\theta + c) = 0$, stating a, b, c.

b. Solve for $0 \leq \theta < 360°$.

7a

7b

8. a. Show $\sin 2\theta + \cos 2\theta = 1$ gives $2\sin\theta(\cos\theta - \sin\theta) = 0$.

b. Solve for $0 \leq \theta < 360°$.

8a

8b

9a. Prove $(\cos 2\theta - \sin 2\theta)^2 = 1 - \sin 4\theta$.

9a

Exercise 7D – Pure Y2 P180-181 33

11a. Show $3\cos^2 x - \sin^2 x = 1 + 2\cos 2x$.

a

b. Sketch $y = 3\cos^2 x - \sin^2 x$, $-\pi \leq x \leq \pi$, labelling intercepts.

b working

b sketch

b reference angles

Extension 34

1a. Use $\sin(A - B)$ to show $\sin(90° - x) = \cos x$.

1b. Solve $2\sin(\theta + 17°) = \cos(\theta + 8°)/\cos(\theta + 17°)$, $0 < \theta < 360°$.

1a

1b

2a. Solve $\sin(\pi/3 - \theta) = \cos\theta/\sqrt{3}$, $0 \leq \theta < 2\pi$.

2b. Find x when $\arcsin(1 - 2x) = \pi/3 - \arcsin x$, $0 < x < 0.5$.

Solutions to question 2 follow on the next slide.

Solution to Extension Question 2 35

2a. Solve $\sin(\pi/3 - \theta) = \cos\theta/\sqrt{3}$, $0 \le \theta < 2\pi$.

2a

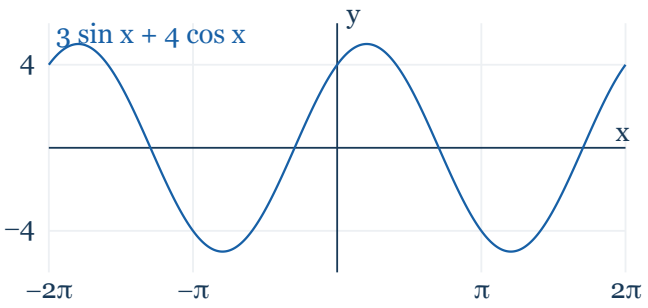
2b. Solve $\arcsin(1 - 2x) = \pi/3 - \arcsin x$, $0 < x < 0.5$.

2b triangle

2b

Combining Sine and Cosine 36

Sketch $y = 3\sin x + 4\cos x$. What do you notice?



It is a sine curve stretched vertically and translated horizontally.

Write $y = R\sin(x + \alpha)$: the amplitude is R, and the horizontal translation is α to the left.

Combining Sine and Cosine 37

Write $3\sin x + 4\cos x$ as $R\sin(x+\alpha)$, giving α in degrees to 1 d.p.

To find R, square the two coefficient equations and add:
$R^2(\cos^2\alpha + \sin^2\alpha) = 3^2 + 4^2$.

1. Expand

2. Compare

3. Amplitude

4. Phase

5. Result

Test Your Understanding 38

Write $\sin x + \cos x = R\sin(x+\alpha)$, giving α in terms of π.

1

Write $\sin x - \sqrt{3}\cos x = R\sin(x-\alpha)$, giving α in terms of π.

2

Further Examples 39

Write $2\cos\theta + 5\sin\theta = R\cos(\theta - \alpha)$, $0 < \alpha < 90^\circ$. Hence solve $2\cos\theta + 5\sin\theta = 3$, $0 < \theta < 360^\circ$.

1

Without calculus, find the maximum of $12\cos\theta + 5\sin\theta$, and the smallest positive θ at which it occurs.

2

Quickfire Maxima 40

Find the maximum and the smallest positive angle (degrees): $20\sin\theta$.

1 maximum

1 angle

Find the maximum and the smallest positive angle (degrees): $5 - 10\sin\theta$.

2 maximum

2 angle

Find the maximum and the smallest positive angle (degrees): $3\cos(\theta + 20^\circ)$.

3 maximum

3 angle

Find the maximum and the smallest positive angle (degrees): $\frac{2}{10 + 3\sin(\theta - 30^\circ)}$.

4 maximum

4 angle

Further Test Your Understanding 41

Edexcel C3 January 2013 Q4.

a. Express $6\cos\theta + 8\sin\theta = R\cos(\theta - \alpha)$, $R > 0, 0 < \alpha < \pi/2$, with α to 3 d.p.

b. For $p(\theta) = 4/(12 + 6\cos\theta + 8\sin\theta)$, $0 \leq \theta \leq 2\pi$, find (i) its maximum, (ii) where it occurs.

a

b(i)

b(ii)

Exercise 7E – Pure Y2 P184-186 42

6. $f(\theta) = \sin\theta + 3\cos\theta = R\sin(\theta + \alpha)$, $R > 0, 0 < \alpha < 90^\circ$.

a. Find R, α.

b. Solve $f(\theta) = 2, 0 \leq \theta < 360^\circ$.

a

b

12. a. Given $\cot\theta + 2 = \operatorname{cosec}\theta$, show $2\sin\theta + \cos\theta = 1$.

b. Solve the original equation, $0 \leq \theta < 360°$.

a

b

Proving Trigonometric Identities 44

Just like Chapter 6 had 'provey' and 'solvey' questions, we also get the 'provey' questions in Chapter 7. Just use the appropriate double angle or addition formula.

Prove $\tan 2\theta = 2/(\cot\theta - \tan\theta)$.

1

Prove $(1 - \cos 2\theta)/\sin 2\theta = \tan\theta$.

2

Test Your Understanding 45

[OCR] Prove $\cot 2x + \operatorname{cosec} 2x = \cot x$.

1

[OCR] By writing $\cos x = \cos(2 \times x/2)$, prove $(1 - \cos x)/(1 + \cos x) = \tan^2(x/2)$.

2

Exercise 7F – Pure Y2 P187-188 46

3a. Show $\tan\theta + \cot\theta = 2\operatorname{cosec} 2\theta$.

b. Hence find $\tan 75° + \cot 75°$.

3a

3b

6. Show $\cos^4\theta = 3/8 + (1/2)\cos 2\theta + (1/8)\cos 4\theta$, with full working.

6

7. Prove
$\sin^2(x+y) - \sin^2(x-y) = \sin 2x \sin 2y.$

7

8. Prove
$\cos 2\theta - \sqrt{3}\sin 2\theta = 2\cos(2\theta + \pi/3).$

8

9. Prove
$4\cos(2\theta - \pi/6)$
$= 2\sqrt{3} - 4\sqrt{3}\sin^2\theta + 4\sin\theta\cos\theta.$

9

10. Show (a) $\cos\theta + \sin\theta = \sqrt{2}\sin(\theta + \pi/4)$,
(b) $\sqrt{3}\sin 2\theta - \cos 2\theta = 2\sin(2\theta - \pi/6).$

10a

10b

Exercise 7F – Pure Y2 P187-188 49

STEP I 2005 Q4.

a. Given $\cos\theta = 3/5$, $3\pi/2 \le \theta \le 2\pi$, show $\sin 2\theta = -24/25$ and evaluate $\cos 3\theta$.

a

b. Prove $\tan 3\theta = (3\tan\theta - \tan^3\theta)/(1 - 3\tan^2\theta)$. Hence find $\tan\theta$ if $\tan 3\theta = 11/2$ and $\pi/4 \le \theta \le \pi/2$.

b

Very Challenging Exam Example 50

Edexcel C3 June 2015 Q8.

a. Prove $\sec 2A + \tan 2A = \frac{\cos A + \sin A}{\cos A - \sin A}$, $A \neq (2n+1)\pi/4$, $n \in \mathbb{Z}$.

Tip: the denominator needs a $\cos A - \sin A$ factor, so use $\cos 2A = \cos^2 A - \sin^2 A$.

Another strategy is to work backwards from the desired result, multiplying by $(\cos A + \sin A)/(\cos A + \sin A)$. This suggests the squared numerator. Present reversible steps and check the factors are nonzero.

a

b. Solve $\sec 2\theta + \tan 2\theta = 1/2$, $0 \le \theta < 2\pi$, to 3 d.p.

b

Modelling 51

June 2013 (withdrawn), Q8.

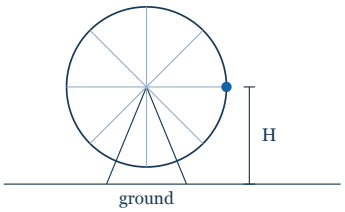
a. Express $9\cos\theta - 2\sin\theta = R\cos(\theta + \alpha)$, $R > 0, 0 < \alpha < \pi/2$, with exact R, α to 4 d.p.

b. State the maximum and the value of θ, $0 < \theta < 2\pi$, where it occurs.

a

b(i)

b(ii)



Ruth models a Ferris-wheel passenger height by $H = 10 - 9\cos(\pi t/5) + 2\sin(\pi t/5)$, metres, with t minutes after starting.

c. Find the maximum height and the first time it occurs (2 d.p.).

d. Find the time for two revolutions.

c

d

Sine and cosine model oscillations such as tides, pendulums and sound waves.

Test Your Understanding 52

June 2010 Q7.

a. Express $2\sin\theta - 1.5\cos\theta = R\sin(\theta - \alpha)$, $R > 0, 0 < \alpha < \pi/2$, with α to 4 d.p.

b. Find the maximum and where it occurs for $0 \le \theta < \pi$.

a

b(i)

b(ii)

Tom models sea height by $H = 6 + 2\sin(4\pi t/25) - 1.5\cos(4\pi t/25)$, $0 \le t < 12$, with t hours after midday.

c. Find the maximum height and when (t to 2 d.p.).

d. Find the times, to the nearest minute, when $H = 7$ m.

c

d

Tip: substitute $\theta = 4\pi t/25$ when using part (a).

6a. Write $65\cos\theta - 20\sin\theta = R\cos(\theta + \alpha)$, $R > 0, 0 < \alpha < \pi/2$, with α to 4 d.p.

A tourist height on a large wheel is $H = 70 - 65\cos 0.2t + 20\sin 0.2t$, metres; t is minutes after boarding, angles in radians.

b. Find the maximum height.

c. Find the time for one revolution.

d. Find the time spent above 100 m per revolution.

a

b

c

d